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Maximize Productivity With AI Tools ✕

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AI models and the training process

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You've explored the way AI models power AI tools. But how exactly are AI models made? And how can understanding the nuts and bolts of a model help you unlock the full potential of AI tools?

Let's find out.

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AI tools and AI models

AI tools and *AI models* are different things, even though they sound similar and are closely related. An **AI tool** is AI-powered software that can automate or assist users with a variety of tasks. An **AI model** is a computer program trained on sets of data to recognize patterns and perform specific tasks. All AI tools are built on top of AI models, which are necessary for AI tools to work.

To better understand the relationship between an AI tool and an AI model, think about a car and its engine.

The car, with its user-friendly parts, like a steering wheel and dashboard, represents an AI tool, but it helps you complete tasks instead of drive from one place to another. And just like cars have different parts, AI tools have different parts, as well. One of the most important parts of a car is its engine. Without it, the car wouldn't work. The same is true for an AI tool, which relies on an AI model to be its "engine," processing the information that you, the user, provide and allowing the AI tool to function.

But not all engines work the same. Similar to how some cars are better for certain things than others— for example, pickup trucks can haul cargo—AI tools are developed for a wide range of applications. There are AI tools for generating text, images, videos, or even computer code. But regardless of the specific function of the AI tool, it needs an AI model to work.

If the car can also drive itself, navigating from one destination to another without you having to adjust the steering wheel, it would operate similar to how an AI agent operates. An **AI agent** is an AI-powered tool that can autonomously perform tasks with little human oversight. For example, AI agents can automatically respond to emails, post content on social media, or monitor computer networks. You set the rules for how the AI agents operate, and the AI agents complete tasks following those rules, allowing you to work on other tasks.

Note: Some AI tools incorporate multiple AI models, working together to achieve more and perform a wider range of tasks. Each model within the tool might be specialized for a specific subtask, which contributes to the overall functionality of the AI tool. You'll explore these types of multimodal tools later in this course.

The process of training AI models

AI designers develop AI models through a process called *training*. Here's an example of the typical steps a designer might take in this process, in this case for building a model that predicts rainfall:

1. **Define the problem to be solved.** AI designers want to predict rain to help people stay dry when commuting to and from work. They consider the AI tool's features and limitations before identifying an AI solution.
2. **Collect relevant data to train the model.** AI designers gather historical data of days when it rained and days when it didn't rain over the past 50 years.
3. **Prepare the data for training.** AI designers prepare the data by labeling important features, like outdoor temperature, humidity, and air pressure, and then noting whether it rained. It's also common to separate the data into two distinct sets: a *training set* to use during the training step and a *validation set* to test after training is complete.
4. **Train the model.** AI designers apply machine learning (ML) programs to the prepared training data. As the ML programs analyze the data, they begin learning how to recognize patterns that indicate the likelihood of rainfall, like the combination of high temperatures, low air pressure, and high humidity.
5. **Evaluate the model.** AI designers use the validation set they prepared earlier to assess their model's ability to predict rainfall accurately and reliably. Analyzing a model's performance can uncover potential issues impacting the model, like insufficient or biased training data. If any issues exist, the AI designers can revisit an earlier step in the process and try a different approach. Once the model performs well with its validation set, the process continues to the next step.
6. **Deploy the model.** When the AI designers are satisfied with their model's performance, they deploy it in an AI tool—helping people in their city stay dry on their way to work.

Model training is an iterative process. AI designers and engineers can repeat each step as many times as necessary and make adjustments until they create the best model possible.

But the process doesn't stop at deployment. Once users interact with a model in practical situations, the model might be exposed to new challenges. AI designers should continuously monitor and collect feedback on their models, ensuring that they continue to perform reliably and to identify areas for improvement. This iterative process of continual refinement makes AI models precise and versatile, resulting in effective, reliable AI tools.

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